

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** 1,2-Dichloroethane

**CAS No** 107-06-2

**Synonyms** Ethylene dichloride; EDC

**Product Code** 167760000; 167760010; 167760025; 167760250; 167765000

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Acute Oral Toxicity  
Acute Inhalation Toxicity - Vapors  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Carcinogenicity  
Specific target organ toxicity - (single exposure)

Category 4  
Category 3  
Category 2  
Category 2  
Category 1B  
Category 3

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Skull and Crossbones



Health Hazard

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H331 - Toxic if inhaled  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H350 - May cause cancer

**Precautionary Statements**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P311 - Call a POISON CENTER or doctor  
P330 - Rinse mouth  
P363 - Wash contaminated clothing before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

**Section 3 - Composition and Information on Ingredients**

| Component           | CAS No   | Weight % |
|---------------------|----------|----------|
| Ethylene dichloride | 107-06-2 | >95      |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.                                                  |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                  |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                     |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                      |
| <b>General Advice</b>                      | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                                                                               |
| <b>Self-Protection of the First Aider</b>  | Use personal protective equipment as required.                                                                                                                                                                                                                                                                                                                                  |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. May cause central nervous system depression: Symptoms may include tightness in the chest, flushing, headache, nausea, vomiting, respiratory depression, weakness, irregular heartbeat, abdominal pain, convulsions, and shock: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed. A patient adversely affected by exposure to this product should not be given adrenaline (epinephrine) or similar heart stimulant since these would increase the risk of cardiac arrhythmias.                                                                                                                                    |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Phosgene, Hydrogen chloride gas.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

| Component           | Australia                                | New Zealand WEL                                 | ACGIH TLV   | The United Kingdom                                                                                                             | Germany |
|---------------------|------------------------------------------|-------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------|---------|
| Ethylene dichloride | TWA: 10 ppm<br>TWA: 40 mg/m <sup>3</sup> | TWA: 5 ppm<br>TWA: 21 mg/m <sup>3</sup><br>Skin | TWA: 10 ppm | STEL: 15 ppm 15 min<br>STEL: 63 mg/m <sup>3</sup> 15 min<br>TWA: 5 ppm 8 hr<br>TWA: 21 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin | Haut    |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

##### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                                                           |
|----------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| Viton (R)      | < 315 minutes     | 0.3 mm          | AS/NZS 2161     | Permeation rate 4 µg/cm <sup>2</sup> /min As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
|                | > 480 minutes     | 0.7 mm          |                 |                                                                                                                          |
| Butyl rubber   | < 70 minutes      | 0.635 mm        |                 |                                                                                                                          |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### Skin and body protection

Long sleeved clothing

##### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

##### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

##### Recommended half mask:-

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

#### Environmental exposure controls

No information available.

## Section 9 - Physical and Chemical Properties

#### Information on basic physical and chemical properties

|                            |                                                 |                                   |
|----------------------------|-------------------------------------------------|-----------------------------------|
| Appearance                 | Colorless                                       |                                   |
| Physical State             | Liquid                                          |                                   |
| Odor                       | sweet                                           |                                   |
| Odor Threshold             | 400 ppm                                         |                                   |
| pH                         | No information available                        |                                   |
| Melting Point/Range        | -35 °C / -31 °F                                 |                                   |
| Softening Point            | No data available                               |                                   |
| Boiling Point/Range        | 81 - 85 °C / 177.8 - 185 °F                     |                                   |
| Flash Point                | 13 °C / 55.4 °F                                 | Method - No information available |
| Evaporation Rate           | 6.5 (Butyl Acetate = 1.0)                       |                                   |
| Flammability (solid,gas)   | Not applicable                                  | Liquid                            |
| Explosion Limits           | <b>Lower</b> 6.2 vol%<br><b>Upper</b> 15.9 vol% |                                   |
| Vapor Pressure             | 65 mmHg @ 29 °C                                 |                                   |
| Vapor Density              | 3.4                                             | (Air = 1.0)                       |
| Specific Gravity / Density | 1.250                                           |                                   |

|                                                |                          |                                             |
|------------------------------------------------|--------------------------|---------------------------------------------|
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                      |
| <b>Water Solubility</b>                        | 8.7 g/L (20°C)           |                                             |
| <b>Solubility in other solvents</b>            | No information available |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                             |
| <b>Component</b>                               | <b>log Pow</b>           |                                             |
| Ethylene dichloride                            | 1.45                     |                                             |
| <b>Autoignition Temperature</b>                | 440 °C / 824 °F          |                                             |
| <b>Decomposition Temperature</b>               | No data available        |                                             |
| <b>Viscosity</b>                               | 0.8 mPa s at 20 °C       |                                             |
| <b>Explosive Properties</b>                    |                          | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available |                                             |
| <b>Other information</b>                       |                          |                                             |
| <b>Molecular Formula</b>                       | C2 H4 Cl2                |                                             |
| <b>Molecular Weight</b>                        | 98.96                    |                                             |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                            |
| <b>Stability</b>                        | Stable under normal conditions.                                                                       |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Bases, Alkali metals.                                                        |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Phosgene. Hydrogen chloride gas.             |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                              |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Category 4                                                       |
| <b>Dermal</b>              | Based on available data, the classification criteria are not met |
| <b>Inhalation</b>          | Category 3                                                       |

| Component           | LD50 Oral                                | LD50 Dermal           | LC50 Inhalation                              |
|---------------------|------------------------------------------|-----------------------|----------------------------------------------|
| Ethylene dichloride | 625 mg/kg ( Rat )<br>413 mg/kg ( Mouse ) | 4890 mg/kg ( Rabbit ) | 28.79 mg/L ( Rat ) 1h<br>7.8 mg/l ( Rat ) 4h |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Based on available data, the classification criteria are not met |

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

(f) carcinogenicity; Category 1B

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component           | Australia | New Zealand | New South Wales | Western Australia | IARC     | EU           | UK | Germany |
|---------------------|-----------|-------------|-----------------|-------------------|----------|--------------|----|---------|
| Ethylene dichloride |           |             |                 |                   | Group 2B | Carc Cat. 1B |    | Cat. 2  |

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs  
Respiratory system  
Central nervous system (CNS)

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met  
Kinematic viscosity:  
> 20.5 mm<sup>2</sup> /s

Symptoms / effects, both acute and delayed May cause central nervous system depression: Symptoms may include tightness in the chest, flushing, headache, nausea, vomiting, respiratory depression, weakness, irregular heartbeat, abdominal pain, convulsions, and shock: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

## Section 12 - Ecological Information

### Ecotoxicity effects

Do not empty into drains. .

| Component           | Freshwater Fish                                                                                                                                                                    | Water Flea                                       | Freshwater Algae                                                                                                  | Microtox |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------|
| Ethylene dichloride | LC50: 230 - 710 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: 110 - 123 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 225 mg/L, 96h static (Oncorhynchus mykiss) | EC50: 140 - 190 mg/L, 48h Static (Daphnia magna) | EC50: > 433 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: = 166 mg/L, 96h static (Desmodesmus subspicatus) |          |

### Persistence and Degradability

Not readily biodegradable

### Persistence

Persistence is unlikely, based on information available.

### Bioaccumulative Potential

Bioaccumulation is unlikely

| Component           | log Pow | Bioconcentration factor (BCF) |
|---------------------|---------|-------------------------------|
| Ethylene dichloride | 1.45    | 2 dimensionless               |

### Mobility

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility Disperses rapidly in air

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

|                               |                                                                                                                                                                                                                                                                                                    |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                           |
| <b>Other Information</b>      | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                     |
|--------------------------------|---------------------|
| <b>UN-No</b>                   | UN1184              |
| <b>Proper Shipping Name</b>    | ETHYLENE DICHLORIDE |
| <b>Hazard Class</b>            | 3                   |
| <b>Subsidiary Hazard Class</b> | 6.1                 |
| <b>Packing Group</b>           | II                  |

### ADG

|                                |                     |
|--------------------------------|---------------------|
| <b>UN-No</b>                   | UN1184              |
| <b>Proper Shipping Name</b>    | ETHYLENE DICHLORIDE |
| <b>Hazard Class</b>            | 3                   |
| <b>Subsidiary Hazard Class</b> | 6.1                 |
| <b>Packing Group</b>           | II                  |

| Component                               | Hazchem Code |
|-----------------------------------------|--------------|
| Ethylene dichloride<br>107-06-2 ( >95 ) | 2YE          |

### IATA

|                                |                     |
|--------------------------------|---------------------|
| <b>UN-No</b>                   | UN1184              |
| <b>Proper Shipping Name</b>    | ETHYLENE DICHLORIDE |
| <b>Hazard Class</b>            | 3                   |
| <b>Subsidiary Hazard Class</b> | 6.1                 |
| <b>Packing Group</b>           | II                  |

|                               |                                 |
|-------------------------------|---------------------------------|
| <b>Environmental hazards</b>  | No hazards identified           |
| <b>Special Precautions</b>    | No special precautions required |
| <b>Additional information</b> | None known                      |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations                      **Australia**

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                      | Standard for the Uniform Scheduling of Medicines and Poisons |
|--------------------------------|--------------------------------------------------------------|
| Ethylene dichloride - 107-06-2 | Schedule 6 listed - present                                  |



## Australian Industrial Chemicals Introduction Scheme (AICIS)

| Component                      | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|--------------------------------|-------------------------------------------------------------|------------------------|
| Ethylene dichloride - 107-06-2 | Present                                                     | -                      |

## Australian - Illicit Drug Precursors/Reagents Substance List

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

## Chemicals of Security Concern

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

## National pollutant inventory Subject to reporting requirements

| Component                      | National pollutant inventory      |
|--------------------------------|-----------------------------------|
| Ethylene dichloride - 107-06-2 | 10 tonne/yr. Threshold category 1 |

## Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

## International Inventories

| Component           | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|---------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Ethylene dichloride | X    | X     | 203-458-1 | -      | X    | X   | -    | X     | X    | X    | X     | KE-10121 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

## Rotterdam Convention (PIC) Chemicals Subject to Prior Informed Consent (PIC)

| Component                      | Rotterdam Convention (PIC) |
|--------------------------------|----------------------------|
| Ethylene dichloride - 107-06-2 | X                          |

## Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component                      | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|--------------------------------|------------------------------------|------------------------------------------------------------------------|
| Ethylene dichloride - 107-06-2 | Annex I - Y45                      | Y45 except substances referenced in Annex I                            |

| Component | CAS No | OECD HPV | Restriction of Hazardous | Seveso III Directive (2012/18/EC) - | Seveso III Directive (2012/18/EC) - |
|-----------|--------|----------|--------------------------|-------------------------------------|-------------------------------------|
|-----------|--------|----------|--------------------------|-------------------------------------|-------------------------------------|

|                     |          |        | Substances (RoHS) | Qualifying Quantities for Major Accident Notification | Qualifying Quantities for Safety Report Requirements |
|---------------------|----------|--------|-------------------|-------------------------------------------------------|------------------------------------------------------|
| Ethylene dichloride | 107-06-2 | Listed | Not applicable    | Not applicable                                        | Not applicable                                       |

## Authorisation/Restrictions according to EU REACH

| Component           | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization                                                          | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                              | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ethylene dichloride | Carcinogenic Category 1B, Article 57<br>Application date: May 22, 2016<br>Sunset date: November 22, 2017<br>Exemption - None | Use restricted. See entry 28.<br>(see link for restriction details)<br>Use restricted. See entry 75.<br>(see link for restriction details) | SVHC Candidate list - Carcinogenic, Article 57a                                                       |

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

## Section 16 - Other Information

## Legend

**AICS** - Australian Inventory of Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

## Revision Date

27-Jun-2024

## Revision Summary

SDS sections updated.

This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of

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**Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).****Disclaimer**

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**End of Safety Data Sheet**